

A Packing of 18 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 18 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.694757. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.686484, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 18 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.694757 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 18$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.686484, obtained by deleting 1 of the 19 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 18/17 = 2.695187$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N18.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 64 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
269475780527720946419141621136763681885243470404918487223139180451037516
313033526129262307587892564679911350349549132453792510854315782857657768
1663540068108590886048299599145314000609636788125
```

denominator:

```
999999999998396400029502361214411107345684664002969378784201952696290624
321845668002223341958668397655847862375298817096049752241910572169788060
338232100208092100448462490453771387809154801919
```

B Coordinates

18 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N18.txt`.

```
35256398098 163049658497 -41993036420 -120618425934515461929783 -496431316279 219312034472 5349926982433669769301 360028852492 -361810961034 22431352783
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